

Started on	Wednesday, 17 September 2025, 14:14
State	Finished
Completed on	Saturday, 20 September 2025, 23:02
Time taken	3 days 8 hours
Grade	11.00 out of 15.00 (73.33%)
Feedback	Congratulations, you've passed! If you want to improve your score, feel free to continue trying to answer the questions you've missed. Your highest score will count as final. Note, getting a higher score will not influence your exam grade, but might help you in a better understanding of the material.

Question 1

Correct

Mark 2.00 out of 2.00

The following message has been encrypted using Caesar's cipher: **byffixulehymmgisfixzlcylhx**. Decrypt the message. Note, the spaces have been removed for better obfuscation. Your answer should have spaces removed as well.

Answer: **Question 2**

Correct

Mark 2.00 out of 2.00

Each month, Alice sends Bob a message, encrypting the name of the city where they should meet using one-time pad. Last month, Eve intercepted an encrypted message "221c15061c0d150c0017", and she knows that Alice then met Bob in **Copenhagen**. The next intercepted message from Alice is 201f001b130b10190c18. In which city do Alice and Bob meet this month? Hint: the messages are encoded in hexadecimal, e.g. "Berlin" becomes "4265726c696e". The answer should be decoded to ACSII, that is, it should be a readable text in English. Following tools can help: <https://gchq.github.io/CyberChef/>, <https://xor.pw>.

Answer: **Question 3**

Incorrect

Mark -2.00 out of 2.00

$\mathcal{H}$ is a hash function that defined as follows: given a number, it returns the sum of all its digits. For example, given 6885 as input, the function returns $6+8+8+5=27$.

Which properties of a cryptographic hash function are ensured?

- ☐ a. Second pre-image resistance
- ☐ b. Easy to compute
- ☐ c. Pre-image resistance
- ☐ d. Collision resistance
- ☒ e. Mapping arbitrary length output to fixed-length input ✖

Question 4

Correct

Mark 3.00 out of 3.00

$N = 39015351411159703781$, $e = 28236791284198039865$ is an RSA public key. Find a secret key d .

Hint: N is small enough to factor using online tools. Once you have the prime factors of N , you can compute d .

Answer: 257



Question 5

Correct

Mark 3.00 out of 3.00

Eve wants to hack into a website by cracking the admin's password. She knows that the website uses 6-digit codes as passwords, and she has found a part of the source code of the website that defines the authentication procedure:

```
function check_credentials($username, $password, $salt) {  
    //hash password  
    $hashed_password = hash('sha256', $password . $salt)  
    // check if credentials exist in database  
    $db = new PDO('sqlite:users.db');  
    $stmt = $db->prepare('SELECT password_hash FROM users WHERE username = :username');  
    $stmt->bindParam(':username', $username); $stmt->execute();  
    $row = $stmt->fetch(PDO::FETCH_ASSOC);  
    // authenticates if the user is found and the stored hashed password matches the hash of the input password  
    if ($row && $row['password_hash'] === $hashed_password)  
    {  
        return true;  
    } else  
    {  
        return false;  
    }  
}
```

Eve has furthermore hacked into the user database of the website, and found that the salt for the administrator's account is `dzr0fv3hz4` and the stored value of the hashed password is `20048def4d4a3922a933a4f84b9cb838c70a8b80402ac45243ed586dd8643bca`. What is the administrator's password?

Hint: you probably would want to write your own script to solve this (you can use any programming language). You also don't have to understand every single line in a code above, it is more important to have a general understanding of what the code does and which parts of it are most highly relevant for your task.

Answer: 558519



Question 6

Correct

Mark 3.00 out of 3.00

You are in charge of an investigation of a possible malware infection. The attached dataset lists all the potentially dangerous files and their attributes (name, size, date added and sha256 hash of the file) on an investigated PC that have been added around the estimated time when the attack has happened. Find out which one of these files contains malware. The answer is the filename of the corresponding file including its extension, e.g. **setup.exe**.

Hint: you can use [ThreatFox dataset](#) of indicators of compromise to identify the file in question.

[files.csv](#)

Answer:

